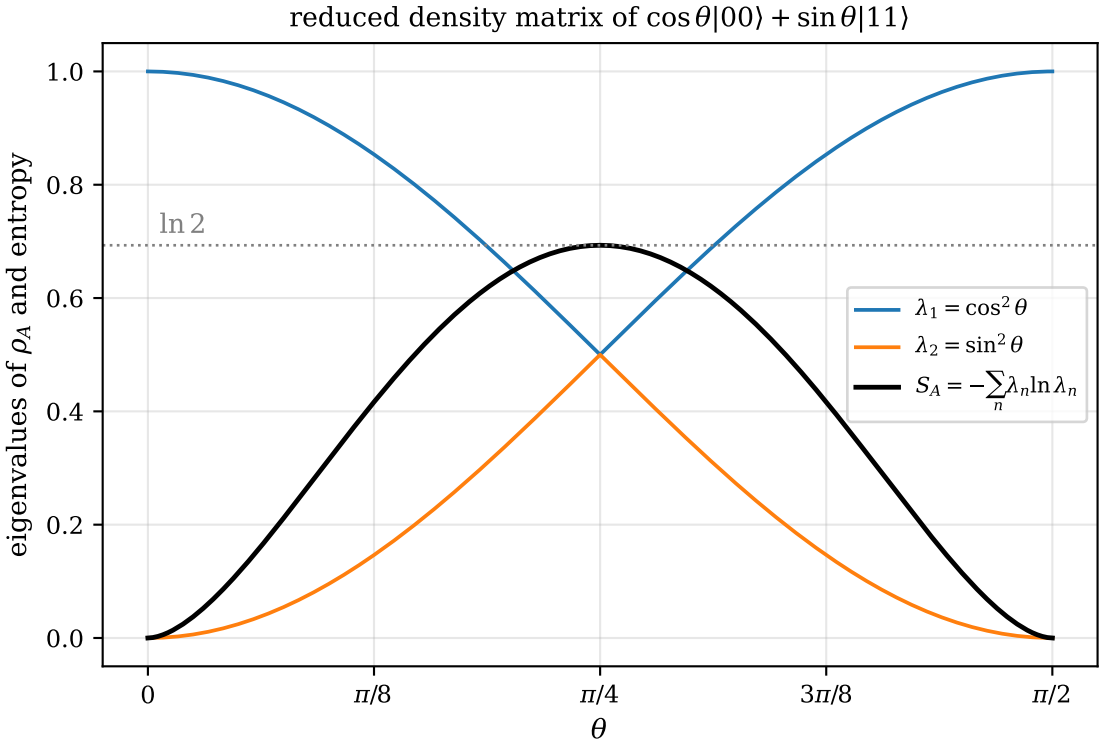
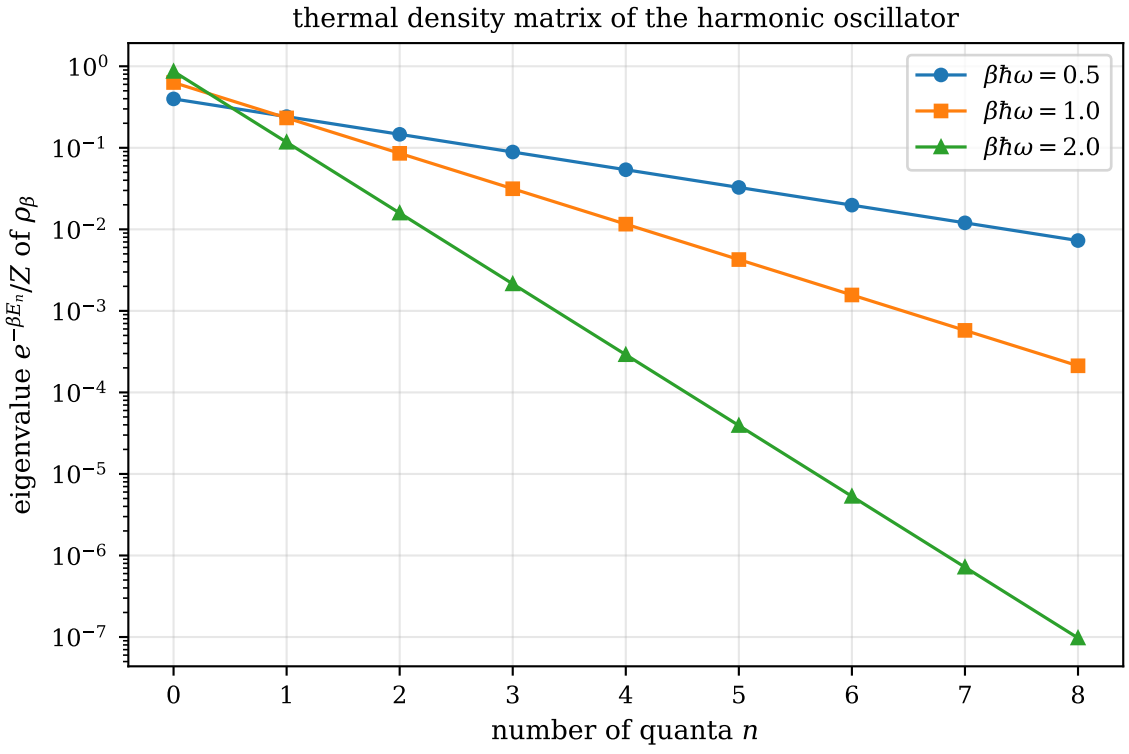

reduced density matrix of $\cos\theta|00\rangle + \sin\theta|11\rangle$
eigenvalues of ρ_A and entropy
$\ln 2$
$\lambda_1 = \cos^2\theta$
$\lambda_2 = \sin^2\theta$
$S_A = -\sum_n \lambda_n \ln\lambda_n$
1.0
0.8
0.6
0.4
0.2
0.0
0
$\pi/8$
$\pi/4$
$3\pi/8$
$\pi/2$
θ


thermal density matrix of the harmonic oscillator
eigenvalue $e^{-\beta E_n}/Z$ of ρ_β
$\beta\hbar\omega = 0.5$
$\beta\hbar\omega = 1.0$
$\beta\hbar\omega = 2.0$
10^{0}
10^{-1}
10^{-2}
10^{-3}
10^{-4}
10^{-5}
10^{-6}
10^{-7}
0
1
2
3
4
5
6
7
8
number of quanta n